

A. Changelog

Here we list changes between this version (<https://arxiv.org/abs/2112.10752v2>) of the paper and the previous version, *i.e.* <https://arxiv.org/abs/2112.10752v1>.

- We updated the results on text-to-image synthesis in Sec. 4.3 which were obtained by training a new, larger model (1.45B parameters). This also includes a new comparison to very recent competing methods on this task that were published on arXiv at the same time as ([59, 109]) or after ([26]) the publication of our work.
- We updated results on class-conditional synthesis on ImageNet in Sec. 4.1, Tab. 3 (see also Sec. D.4) obtained by retraining the model with a larger batch size. The corresponding qualitative results in Fig. 26 and Fig. 27 were also updated. Both the updated text-to-image and the class-conditional model now use classifier-free guidance [32] as a measure to increase visual fidelity.
- We conducted a user study (following the scheme suggested by Saharia et al [72]) which provides additional evaluation for our inpainting (Sec. 4.5) and superresolution models (Sec. 4.4).
- Added Fig. 5 to the main paper, moved Fig. 18 to the appendix, added Fig. 13 to the appendix.

B. Detailed Information on Denoising Diffusion Models

Diffusion models can be specified in terms of a signal-to-noise ratio $\text{SNR}(t) = \frac{\alpha_t^2}{\sigma_t^2}$ consisting of sequences $(\alpha_t)_{t=1}^T$ and $(\sigma_t)_{t=1}^T$ which, starting from a data sample x_0 , define a forward diffusion process q as

$$q(x_t|x_0) = \mathcal{N}(x_t|\alpha_t x_0, \sigma_t^2 \mathbb{I}) \quad (4)$$

with the Markov structure for $s < t$:

$$q(x_t|x_s) = \mathcal{N}(x_t|\alpha_{t|s} x_s, \sigma_{t|s}^2 \mathbb{I}) \quad (5)$$

$$\alpha_{t|s} = \frac{\alpha_t}{\alpha_s} \quad (6)$$

$$\sigma_{t|s}^2 = \sigma_t^2 - \alpha_{t|s}^2 \sigma_s^2 \quad (7)$$

Denoising diffusion models are generative models $p(x_0)$ which revert this process with a similar Markov structure running backward in time, *i.e.* they are specified as

$$p(x_0) = \int_z p(x_T) \prod_{t=1}^T p(x_{t-1}|x_t) \quad (8)$$

The evidence lower bound (ELBO) associated with this model then decomposes over the discrete time steps as

$$-\log p(x_0) \leq \mathbb{KL}(q(x_T|x_0)|p(x_T)) + \sum_{t=1}^T \mathbb{E}_{q(x_t|x_0)} \mathbb{KL}(q(x_{t-1}|x_t, x_0)|p(x_{t-1}|x_t)) \quad (9)$$

The prior $p(x_T)$ is typically chosen as a standard normal distribution and the first term of the ELBO then depends only on the final signal-to-noise ratio $\text{SNR}(T)$. To minimize the remaining terms, a common choice to parameterize $p(x_{t-1}|x_t)$ is to specify it in terms of the true posterior $q(x_{t-1}|x_t, x_0)$ but with the unknown x_0 replaced by an estimate $x_\theta(x_t, t)$ based on the current step x_t . This gives [45]

$$p(x_{t-1}|x_t) := q(x_{t-1}|x_t, x_\theta(x_t, t)) \quad (10)$$

$$= \mathcal{N}(x_{t-1}|\mu_\theta(x_t, t), \sigma_{t|t-1}^2 \frac{\sigma_{t-1}^2}{\sigma_t^2} \mathbb{I}), \quad (11)$$

where the mean can be expressed as

$$\mu_\theta(x_t, t) = \frac{\alpha_{t|t-1} \sigma_{t-1}^2}{\sigma_t^2} x_t + \frac{\alpha_{t-1} \sigma_{t|t-1}^2}{\sigma_t^2} x_\theta(x_t, t). \quad (12)$$

	<i>LDM-1</i>	<i>LDM-2</i>	<i>LDM-4</i>	<i>LDM-8</i>	<i>LDM-16</i>	<i>LDM-32</i>
<i>z</i> -shape	$256 \times 256 \times 3$	$128 \times 128 \times 2$	$64 \times 64 \times 3$	$32 \times 32 \times 4$	$16 \times 16 \times 8$	$88 \times 8 \times 32$
$ \mathcal{Z} $	-	2048	8192	16384	16384	16384
Diffusion steps	1000	1000	1000	1000	1000	1000
Noise Schedule	linear	linear	linear	linear	linear	linear
Model Size	270M	265M	274M	258M	260M	258M
Channels	192	192	224	256	256	256
Depth	2	2	2	2	2	2
Channel Multiplier	1,1,2,2,4,4	1,2,2,4,4	1,2,3,4	1,2,4	1,2,4	1,2,4
Attention resolutions	32, 16, 8	32, 16, 8	32, 16, 8	32, 16, 8	16, 8, 4	8, 4, 2
Head Channels	32	32	32	32	32	32
Batch Size	9	11	48	96	128	128
Iterations*	500k	500k	500k	500k	500k	500k
Learning Rate	9e-5	1.1e-4	9.6e-5	9.6e-5	1.3e-4	1.3e-4

Table 14. Hyperparameters for the unconditional *LDMs* trained on the CelebA dataset for the analysis in Fig. 7. All models trained on a single NVIDIA A100. *: All models are trained for 500k iterations. If converging earlier, we used the best checkpoint for assessing the provided FID scores.

Task	Text-to-Image	Layout-to-Image		Class-Label-to-Image	Super Resolution	Inpainting	Semantic-Map-to-Image
Dataset	LAION	OpenImages	COCO	ImageNet	ImageNet	Places	Landscapes
f	8	4	8	4	4	4	8
z -shape	$32 \times 32 \times 4$	$64 \times 64 \times 3$	$32 \times 32 \times 4$	$64 \times 64 \times 3$	$64 \times 64 \times 3$	$64 \times 64 \times 3$	$32 \times 32 \times 4$
$ \mathcal{Z} $	-	8192	16384	8192	8192	8192	16384
Diffusion steps	1000	1000	1000	1000	1000	1000	1000
Noise Schedule	linear	linear	linear	linear	linear	linear	linear
Model Size	1.45B	306M	345M	395M	169M	215M	215M
Channels	320	128	192	192	160	128	128
Depth	2	2	2	2	2	2	2
Channel Multiplier	1,2,4,4	1,2,3,4	1,2,4	1,2,3,5	1,2,2,4	1,4,8	1,4,8
Number of Heads	8	1	1	1	1	1	1
Dropout	-	-	0.1	-	-	-	-
Batch Size	680	24	48	1200	64	128	48
Iterations	390K	4.4M	170K	178K	860K	360K	360K
Learning Rate	1.0e-4	4.8e-5	4.8e-5	1.0e-4	6.4e-5	1.0e-6	4.8e-5
Conditioning	CA	CA	CA	CA	concat	concat	concat
(C)A-resolutions	32, 16, 8	32, 16, 8	32, 16, 8	32, 16, 8	-	-	-
Embedding Dimension	1280	512	512	512	-	-	-
Transformer Depth	1	3	2	1	-	-	-

Table 15. Hyperparameters for the conditional *LDMs* from Sec. 4. All models trained on a single NVIDIA A100 except for the inpainting model which was trained on eight V100.

$$\zeta \leftarrow \text{TokEmb}(y) + \text{PosEmb}(y) \quad (18)$$

for $i = 1, \dots, N$:

$$\zeta_1 \leftarrow \text{LayerNorm}(\zeta) \quad (19)$$

$$\zeta_2 \leftarrow \text{MultiHeadSelfAttention}(\zeta_1) + \zeta \quad (20)$$

$$\zeta_3 \leftarrow \text{LayerNorm}(\zeta_2) \quad (21)$$

$$\zeta \leftarrow \text{MLP}(\zeta_3) + \zeta_2 \quad (22)$$

$$\zeta \leftarrow \text{LayerNorm}(\zeta) \quad (23)$$

$$(24)$$

With ζ available, the conditioning is mapped into the UNet via the cross-attention mechanism as depicted in Fig. 3. We modify the “ablated UNet” [15] architecture and replace the self-attention layer with a shallow (unmasked) transformer consisting of T blocks with alternating layers of (i) self-attention, (ii) a position-wise MLP and (iii) a cross-attention layer;

G. Details on Autoencoder Models

We train all our autoencoder models in an adversarial manner following [23], such that a patch-based discriminator D_ψ is optimized to differentiate original images from reconstructions $\mathcal{D}(\mathcal{E}(x))$. To avoid arbitrarily scaled latent spaces, we regularize the latent z to be zero centered and obtain small variance by introducing a regularizing loss term L_{reg} .

We investigate two different regularization methods: (i) a low-weighted Kullback-Leibler-term between $q_{\mathcal{E}}(z|x) = \mathcal{N}(z; \mathcal{E}_\mu, \mathcal{E}_{\sigma^2})$ and a standard normal distribution $\mathcal{N}(z; 0, 1)$ as in a standard variational autoencoder [46, 69], and, (ii) regularizing the latent space with a vector quantization layer by learning a codebook of $|\mathcal{Z}|$ different exemplars [96].

To obtain high-fidelity reconstructions we only use a very small regularization for both scenarios, *i.e.* we either weight the $\mathbb{KL}$ term by a factor $\sim 10^{-6}$ or choose a high codebook dimensionality $|\mathcal{Z}|$.

The full objective to train the autoencoding model $(\mathcal{E}, \mathcal{D})$ reads:

$$L_{\text{Autoencoder}} = \min_{\mathcal{E}, \mathcal{D}} \max_{\psi} \left(L_{rec}(x, \mathcal{D}(\mathcal{E}(x))) - L_{adv}(\mathcal{D}(\mathcal{E}(x))) + \log D_\psi(x) + L_{reg}(x; \mathcal{E}, \mathcal{D}) \right) \quad (25)$$

DM Training in Latent Space Note that for training diffusion models on the learned latent space, we again distinguish two cases when learning $p(z)$ or $p(z|y)$ (Sec. 4.3): (i) For a KL-regularized latent space, we sample $z = \mathcal{E}_\mu(x) + \mathcal{E}_\sigma(x) \cdot \varepsilon =: \mathcal{E}(x)$, where $\varepsilon \sim \mathcal{N}(0, 1)$. When rescaling the latent, we estimate the component-wise variance

$$\hat{\sigma}^2 = \frac{1}{bchw} \sum_{b,c,h,w} (z^{b,c,h,w} - \hat{\mu})^2$$

from the first batch in the data, where $\hat{\mu} = \frac{1}{bchw} \sum_{b,c,h,w} z^{b,c,h,w}$. The output of $\mathcal{E}$ is scaled such that the rescaled latent has unit standard deviation, *i.e.* $z \leftarrow \frac{z}{\hat{\sigma}} = \frac{\mathcal{E}(x)}{\hat{\sigma}}$. (ii) For a VQ-regularized latent space, we extract z *before* the quantization layer and absorb the quantization operation into the decoder, *i.e.* it can be interpreted as the first layer of $\mathcal{D}$.

H. Additional Qualitative Results

Finally, we provide additional qualitative results for our landscapes model (Fig. 12, 23, 24 and 25), our class-conditional ImageNet model (Fig. 26 - 27) and our unconditional models for the CelebA-HQ, FFHQ and LSUN datasets (Fig. 28 - 31). Similar as for the inpainting model in Sec. 4.5 we also fine-tuned the semantic landscapes model from Sec. 4.3.2 directly on 512² images and depict qualitative results in Fig. 12 and Fig. 23. For our those models trained on comparably small datasets, we additionally show nearest neighbors in VGG [79] feature space for samples from our models in Fig. 32 - 34.